

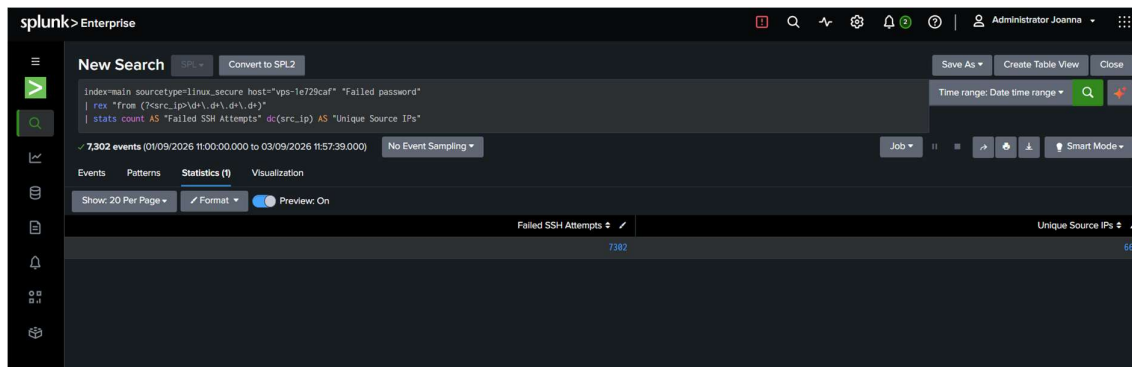
Joanna Bizior

03.09.2026

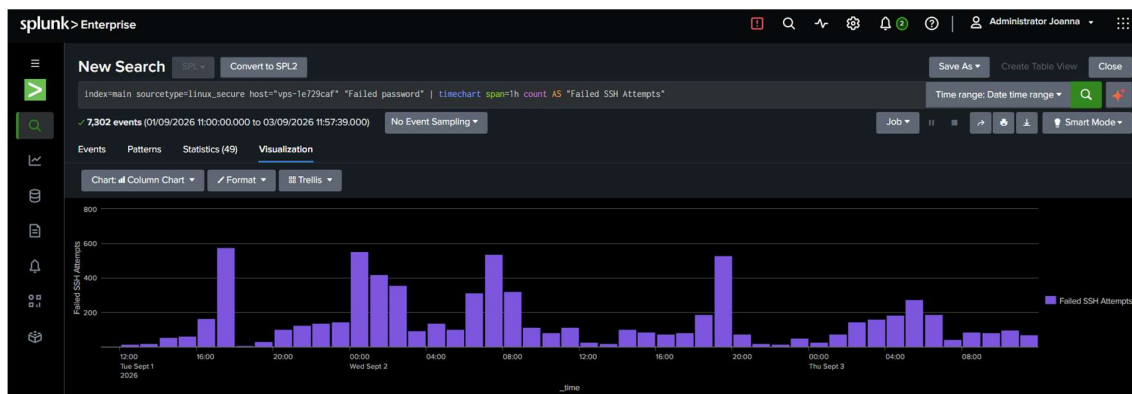
The purpose of this lab is to investigate the SSH activity on my public VPS and see how often it is being targeted by failed login attempts. I will use Splunk to analyse the authentication logs, identify suspicious activity and create a detection for possible SSH brute-force attacks. After collecting the initial results, I will disable SSH password authentication and use SSH keys only. The same activity will then be checked again after 48 hours to see what difference this security change makes.

Initial investigation, before SSH hardening

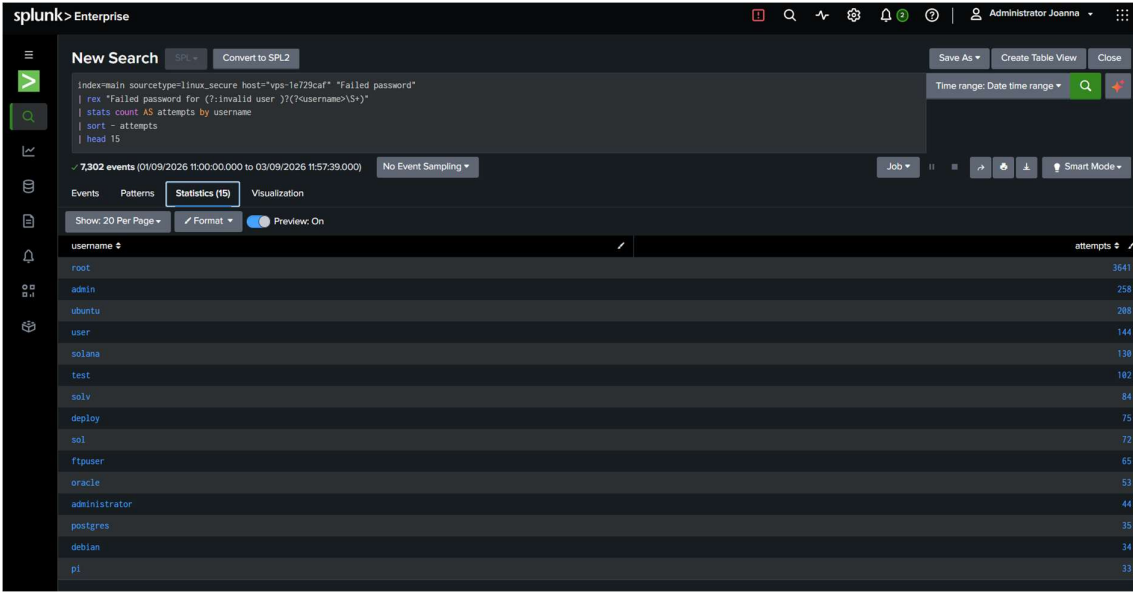
A Splunk search was used to identify failed SSH password attempts on the public-facing VPS between 1 and 3 September. The results showed **7,302 failed login attempts from 66 unique source IP addresses**, demonstrating how quickly a publicly accessible server can become a target for automated login attempts.



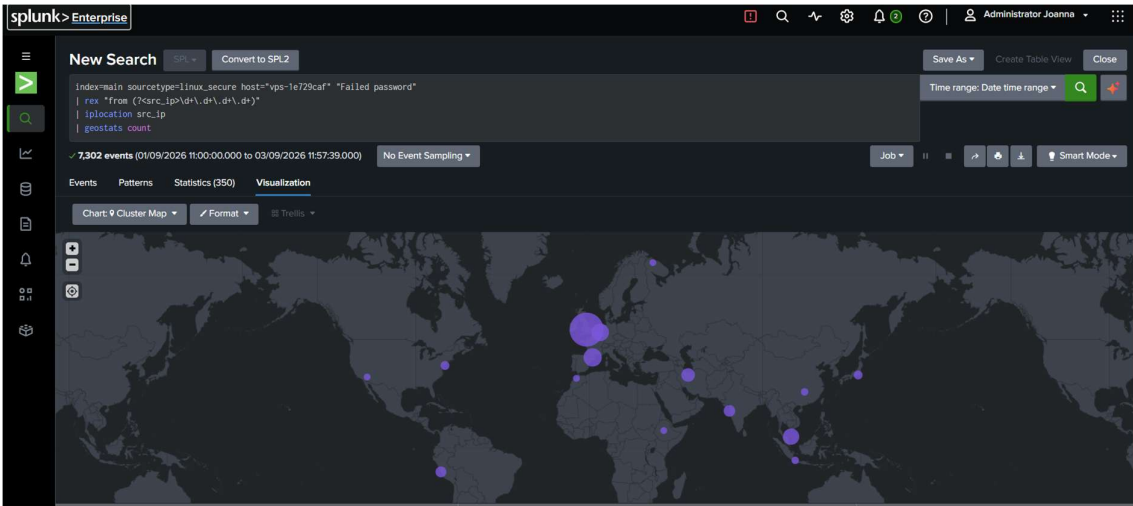
Next, I grouped the failed SSH attempts into one-hour intervals to see how the activity changed over time. I observed continuous login attempts throughout the monitoring period, with several spikes reaching over 500 attempts within a single hour.



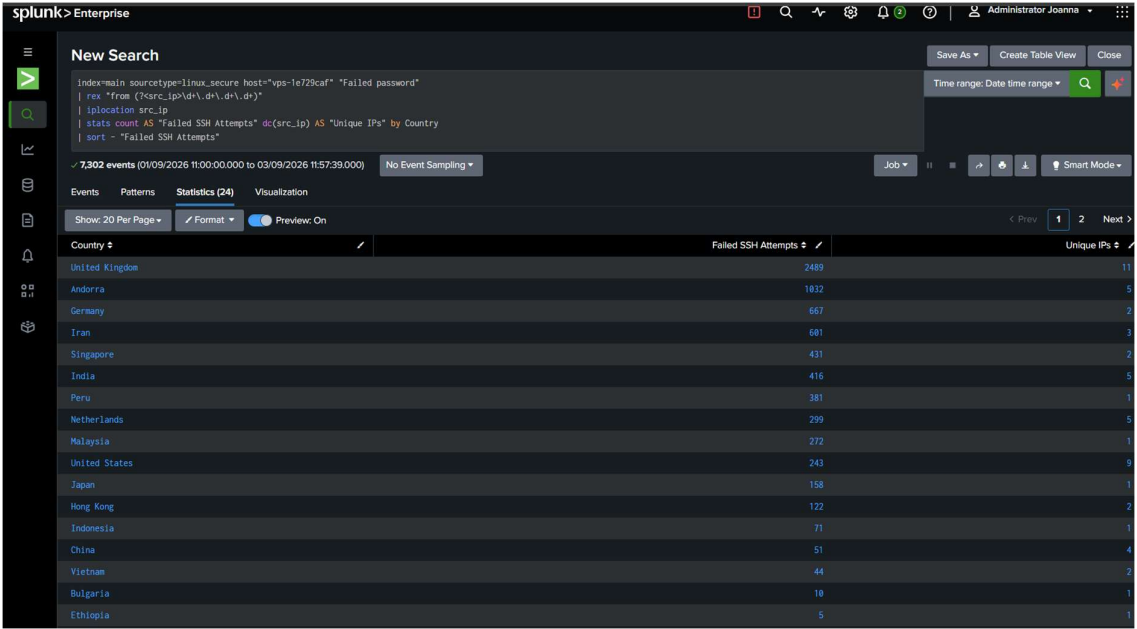
Next, I analysed which usernames were targeted most frequently during the failed SSH login attempts. The results show that root was by far the most targeted account with 3,641 attempts, followed by common usernames such as admin, ubuntu and user.



In this step, I used IP geolocation to visualise where the failed SSH attempts were coming from. The map shows that the VPS was receiving login attempts from IP addresses located across multiple regions worldwide, showing the global nature of automated SSH scanning and brute-force activity.



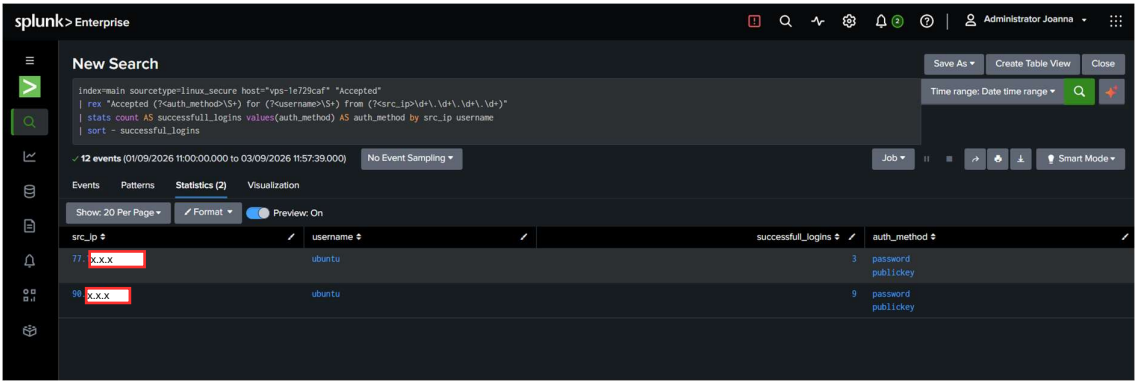
In this step, I grouped the failed SSH attempts by country to get a clearer view of where the activity was originating from. The highest number of attempts came from IP addresses geolocated to the United Kingdom, Andorra and Germany, with activity also observed across many other countries.



The screenshot shows a Splunk Enterprise search interface. The search bar contains the following query: `index=main sourcetype=linux_secure host="vps-1e729caf" "Failed password" | rex "from (?<src_ip>[d+\.d+\.d+\.d+])" | iplocation src_ip | stats count AS "Failed SSH Attempts" dc(src_ip) AS "Unique IPs" by Country | sort - "Failed SSH Attempts"`. The results are displayed in a table with 24 statistics. The table has three columns: Country, Failed SSH Attempts, and Unique IPs. The data is sorted by Failed SSH Attempts in descending order.

Country	Failed SSH Attempts	Unique IPs
United Kingdom	2489	11
Andorra	1832	5
Germany	667	2
Iran	601	3
Singapore	431	2
India	416	5
Peru	381	1
Netherlands	299	5
Malaysia	272	1
United States	243	9
Japan	158	1
Hong Kong	122	2
Indonesia	71	1
China	51	4
Vietnam	44	2
Bulgaria	18	1
Ethiopia	5	1

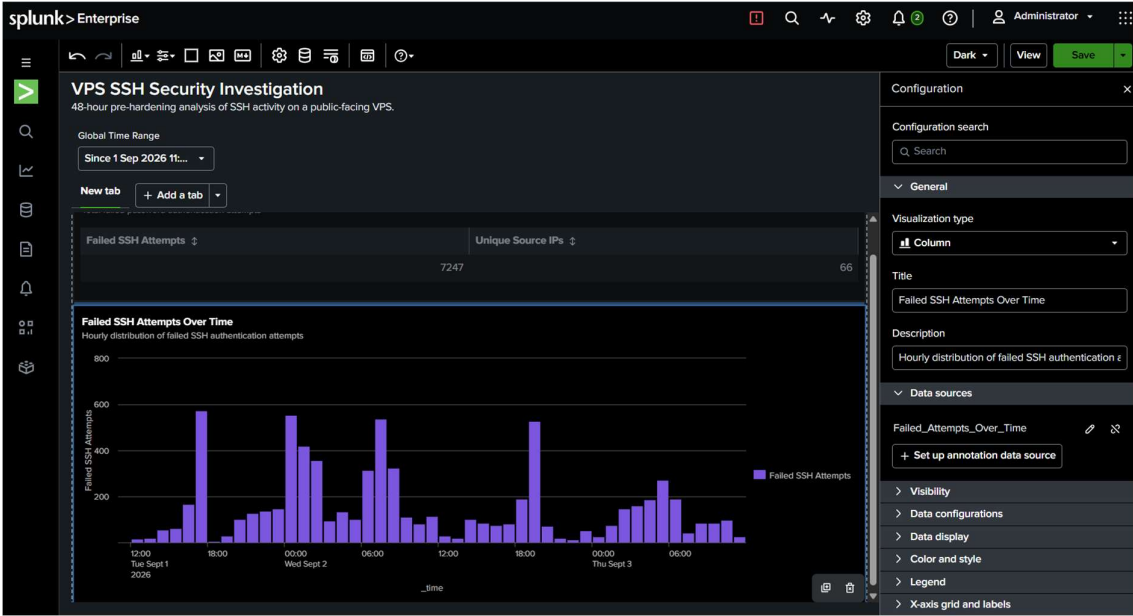
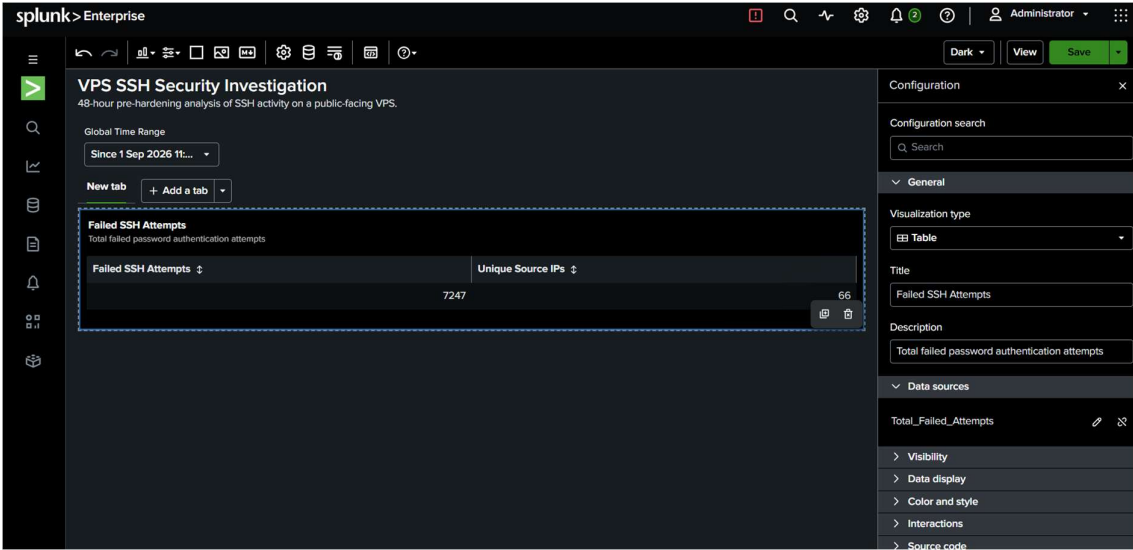
Finally, I checked the logs for successful SSH authentications to make sure the failed attempts had not resulted in unauthorised access. I identified 12 successful logins from two known IP addresses, with both password and public-key authentication recorded. Both of these addresses belong to myself, and another authorised administrator. No successful logins from the suspicious source IPs identified earlier were observed.

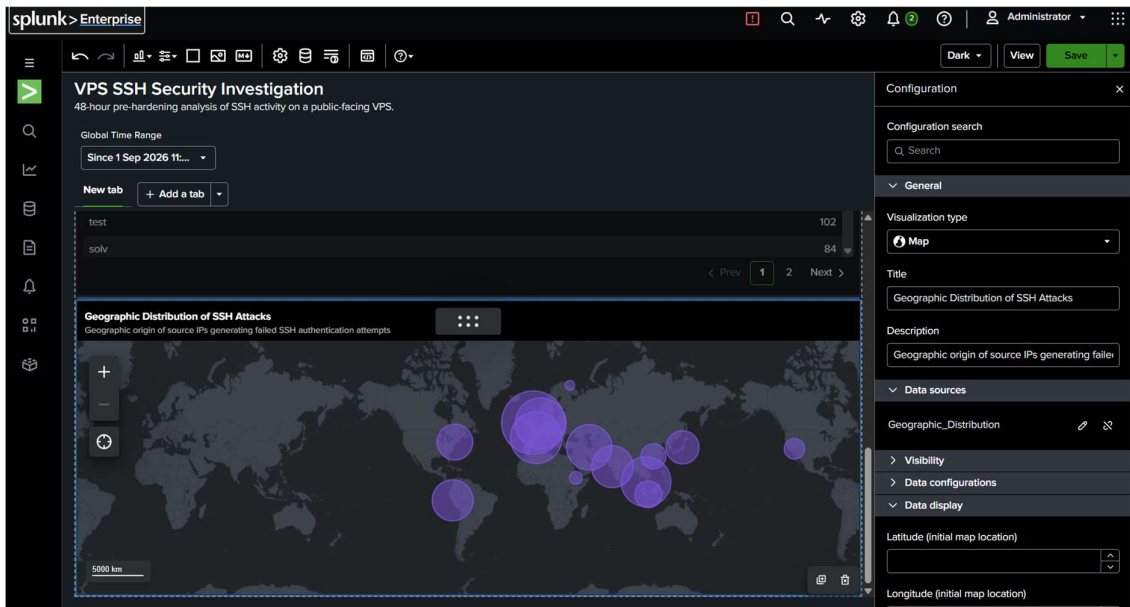
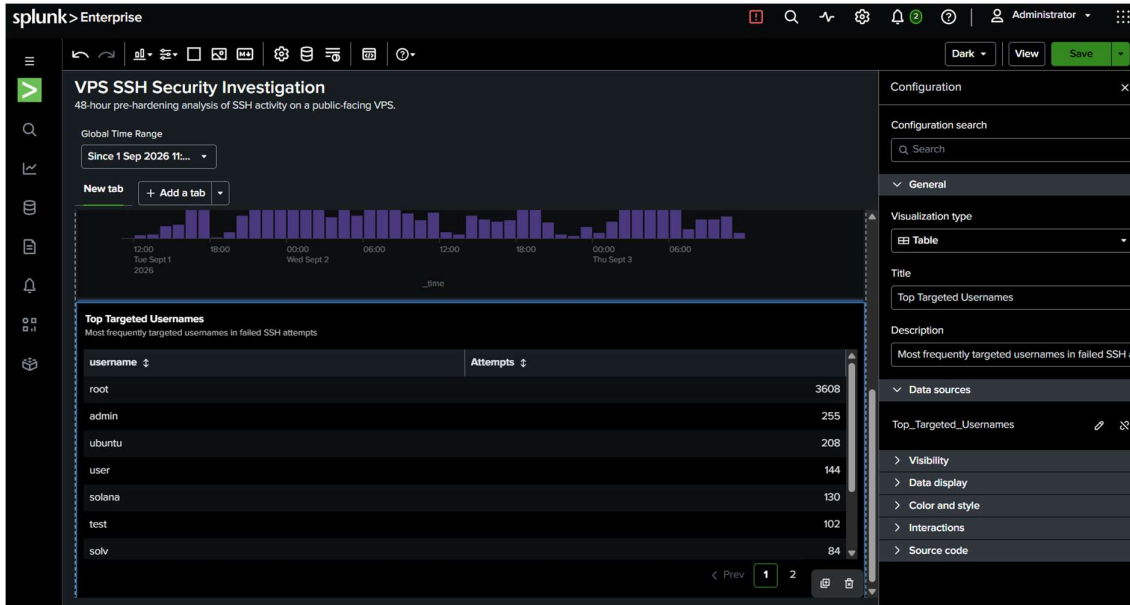


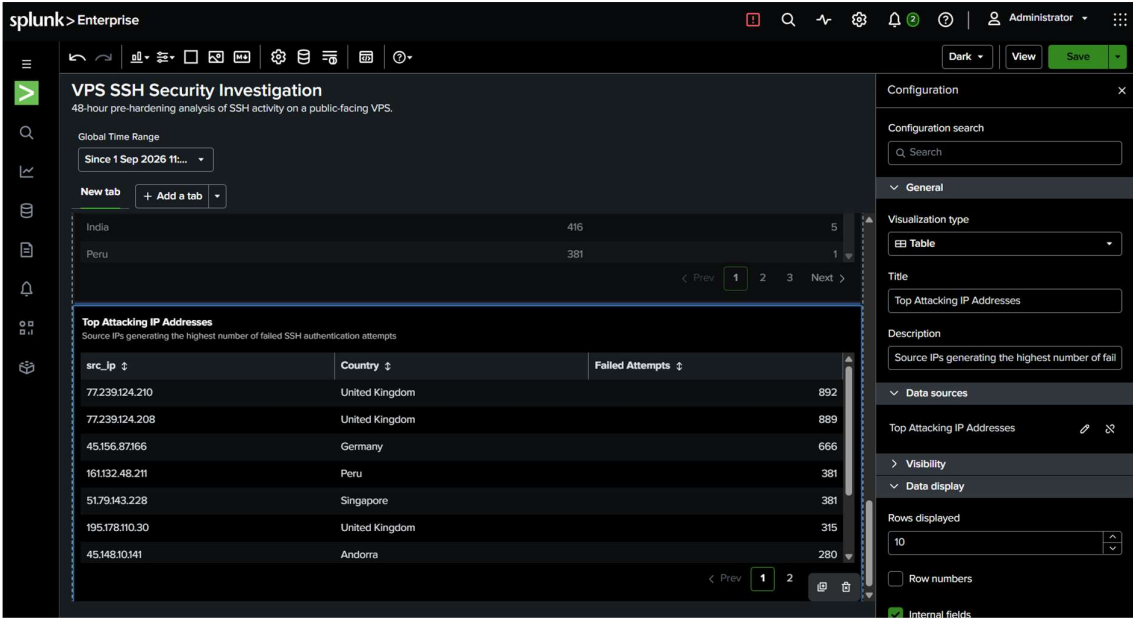
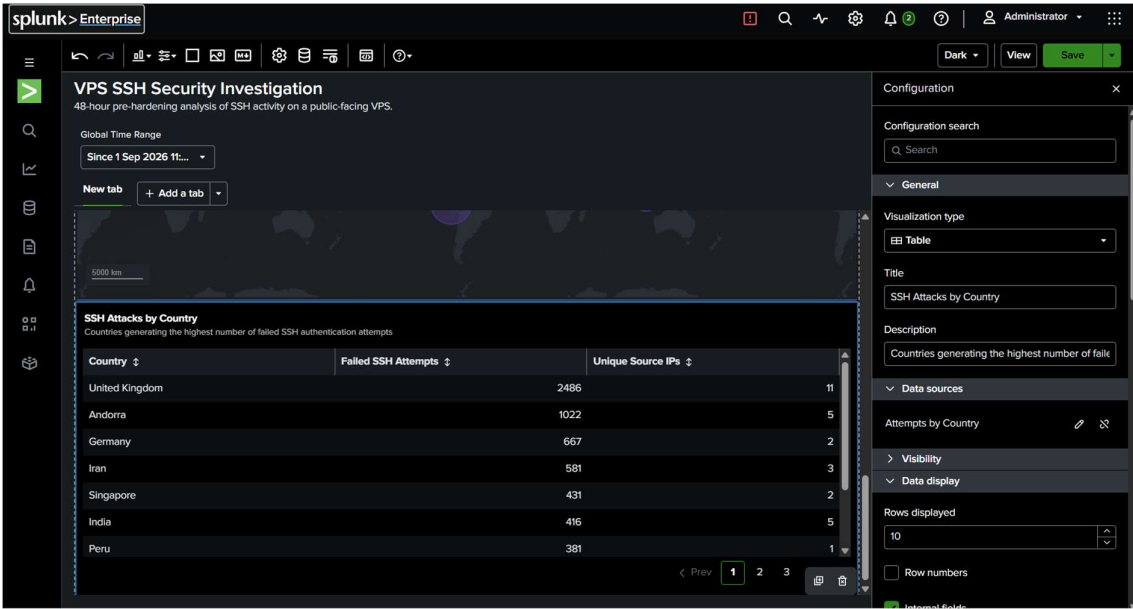
The screenshot shows a Splunk Enterprise search interface. The search bar contains the following query: `index=main sourcetype=linux_secure host="vps-1e729caf" "Accepted" | rex "Accepted (?<auth_method>[a-zA-Z]+) for (?<username>[a-zA-Z]+) from (?<src_ip>[d+\.d+\.d+\.d+])" | stats count AS successful_logins values(auth_method) AS auth_method by src_ip username | sort - successful_logins`. The results are displayed in a table with 2 statistics. The table has four columns: src_ip, username, successful_logins, and auth_method. The data is sorted by successful_logins in descending order.

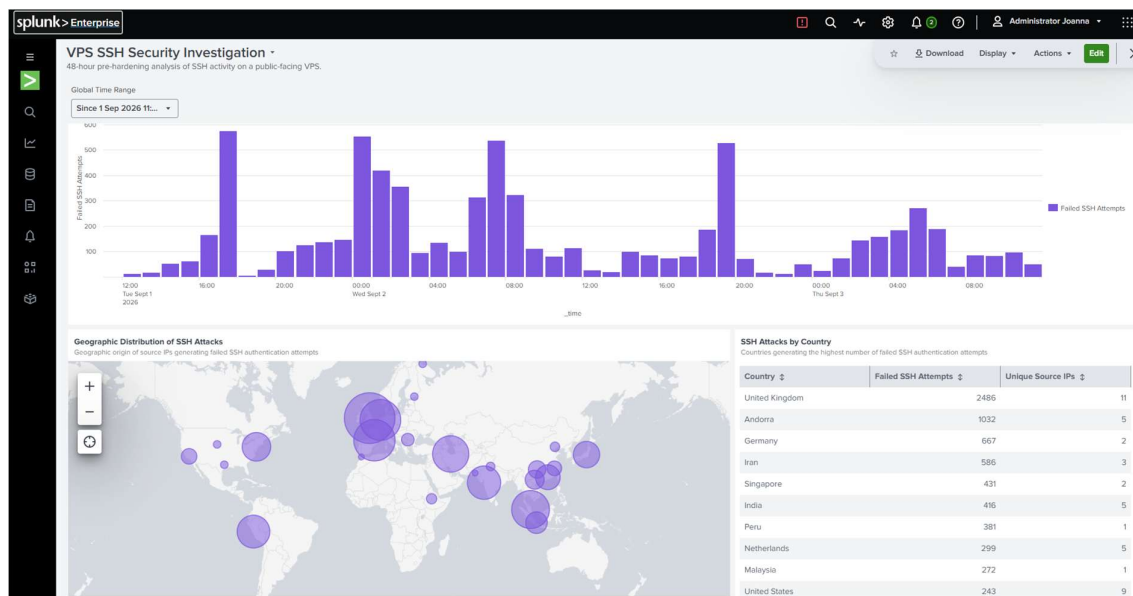
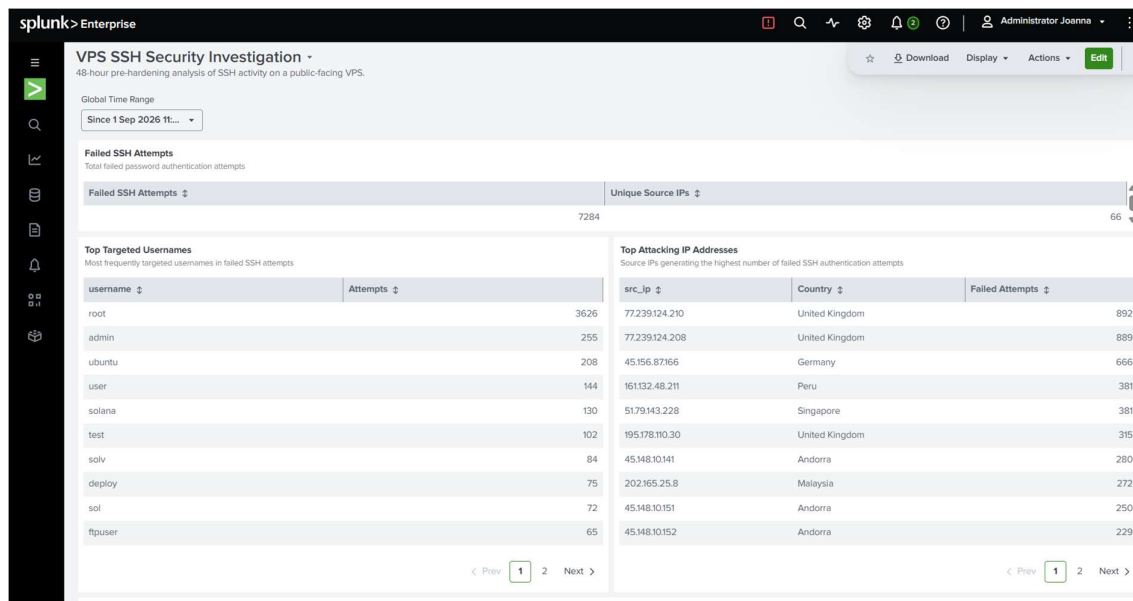
src_ip	username	successful_logins	auth_method
77.X.X.X	ubuntu	3	password publickey
98.X.X.X	ubuntu	9	password publickey

To visualise all of the findings in one place, I created a dedicated Splunk dashboard combining the key results from my investigation. The screenshots below show the process of building the dashboard and bringing the different searches and visualisations together.









Disabling SSH password logins

In the first step, I verify that I can log in to the VPS using my public key:

```
PS C:\WINDOWS\System32> ssh -i $env:USERPROFILE\.ssh\luanti_vps ubuntu@51.195.218.73
Enter passphrase for key 'C:\Users\asia1\.ssh\luanti_vps':
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)
```

```
Last login: Wed Sep  2 20:09:35 2026 from 90.X.X.X
ubuntu@vps-1e729caf:~$
```


Next, I verify what is the current authentication configuration:

```
ubuntu@vps-1e729caf:~$ sudo sshd -T | grep -E 'passwordauthentication|pubkeyauthentication|permitrootlogin|kbdinteractiveauthentication'
permitrootlogin prohibit-password
pubkeyauthentication yes
passwordauthentication yes
kbdinteractiveauthentication no
ubuntu@vps-1e729caf:~$
```

As we can see above, root password login is already disabled, now I will also disable password authentication.

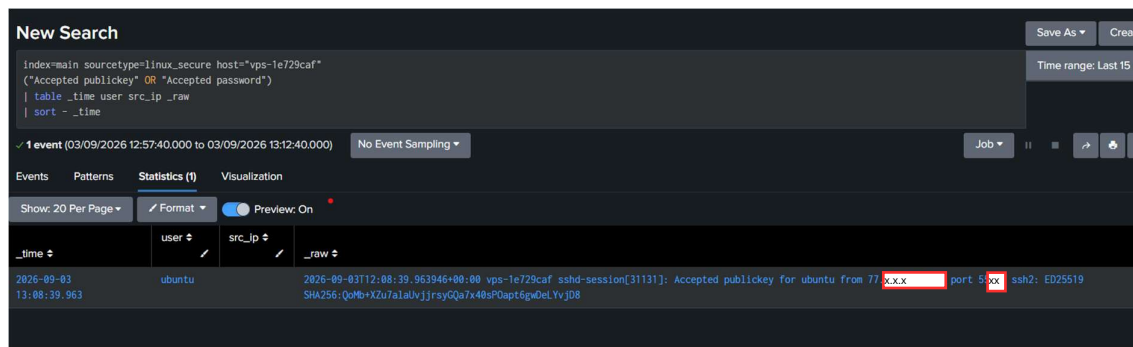
I used grep to search the SSH configuration files and found that 50-cloud-init.conf contained PasswordAuthentication yes. I then changed this setting to no using sed, validated the configuration with sshd -t, and reloaded the SSH service. Finally, sshd -T confirmed that password authentication was successfully disabled.

```
ubuntu@vps-1e729caf:~$ echo 'PasswordAuthentication no' | sudo tee /etc/ssh/sshd_config.d/99-key-only.conf
PasswordAuthentication no
ubuntu@vps-1e729caf:~$ sudo sshd -t
ubuntu@vps-1e729caf:~$ sudo systemctl reload ssh
ubuntu@vps-1e729caf:~$ sudo sshd -T | grep passwordauthentication
passwordauthentication yes
ubuntu@vps-1e729caf:~$ sudo sshd -T | grep passwordauthentication
passwordauthentication yes
ubuntu@vps-1e729caf:~$ sudo systemctl reload ssh
ubuntu@vps-1e729caf:~$ sudo sshd -T | grep passwordauthentication
passwordauthentication yes
ubuntu@vps-1e729caf:~$ sudo grep -Rni "PasswordAuthentication" /etc/ssh/sshd_config /etc/ssh/sshd_config.d/
/etc/ssh/sshd_config:78:#PasswordAuthentication yes
/etc/ssh/sshd_config:102:# PasswordAuthentication. Depending on your PAM configuration,
/etc/ssh/sshd_config:106:# PAM authentication, then enable this but set PasswordAuthentication
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf:1:PasswordAuthentication no
/etc/ssh/sshd_config.d/50-cloud-init.conf:1:PasswordAuthentication yes
/etc/ssh/sshd_config.d/99-key-only.conf:1:PasswordAuthentication no
ubuntu@vps-1e729caf:~$ sudo sed -i 's/'PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config.d/50-cloud-init.conf
ubuntu@vps-1e729caf:~$ sudo sshd -t
ubuntu@vps-1e729caf:~$ sudo systemctl reload ssh
ubuntu@vps-1e729caf:~$ sudo sshd -T | grep passwordauthentication
passwordauthentication no
ubuntu@vps-1e729caf:~$
```

To confirm that I can still securely login to the vps, I opened another session:

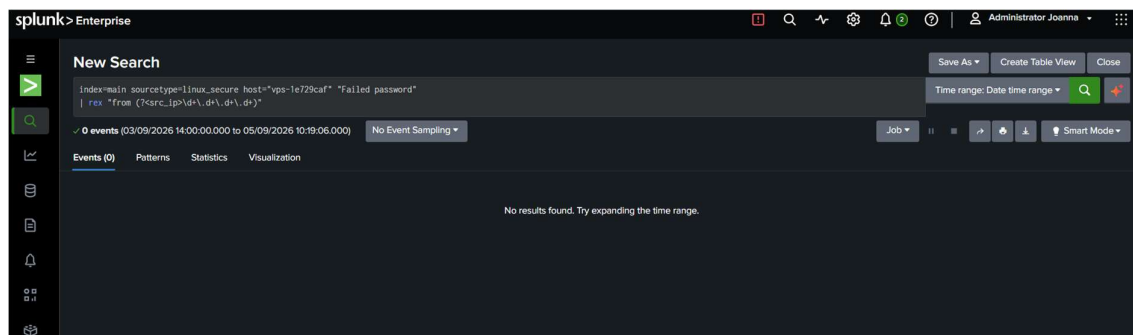
```
Last login: Thu Sep  3 11:55:32 2026 from 77.X.X.X
ubuntu@vps-1e729caf:~$
```

Finally, I verified the SSH hardening from the Splunk logs. After disabling password-based authentication, I opened a new SSH session to the VPS and confirmed that the login was recorded as Accepted publickey. This confirmed that key-based authentication remained functional after the configuration change.



Investigation post SSH hardening

After disabling password-based SSH authentication, I searched the VPS authentication logs for new Failed password events. No events were recorded during the selected post-hardening period, confirming that password-based authentication was no longer available.



Although no further Failed password events were recorded after password authentication was disabled, the server continued to receive a significant amount of automated SSH activity. During the same post-hardening period, Splunk recorded 4,140 events related to invalid usernames and pre-authentication connections. This shows that SSH hardening successfully removed password-based authentication, but did not stop automated bots from scanning and attempting to interact with the publicly exposed SSH service.

splunk> Enterprise

New Search

```
index=main sourcetype=linux_secure host=vps-le729caf
("invalid user" OR "preauth")
| table _time _raw
| sort - _time
```

✓ 4,140 events (03/09/2026 14:00:00.000 to 05/09/2026 10:10:00.000) No Event Sampling ▾

Events Patterns **Statistics (4,140)** Visualization

Show: 20 Per Page ▾ Format ▾ Preview: On

Time range: Date time range ▾ 🔍 ⚙️

Job ▮ = ↗️ 📄 👤 Smart Mode ▾

_time ↓	_raw ↑
2026-09-04 23:08:18.769	2026-09-04T22:08:18.769546+00:00 vps-le729caf sshd-session(42202): Connection closed by invalid user solana from 195.178.110.30 port 52972 [preauth]
2026-09-04 23:08:18.734	2026-09-04T22:08:18.734690+00:00 vps-le729caf sshd-session(42202): Invalid user solana from 195.178.110.30 port 52972
2026-09-04 23:06:35.605	2026-09-04T22:06:35.605213+00:00 vps-le729caf sshd-session(42199): Connection closed by authenticating user root 138.226.239.234 port 31358 [preauth]
2026-09-04 23:05:50.327	2026-09-04T22:05:50.327083+00:00 vps-le729caf sshd-session(42196): Connection closed by invalid user pul 66.93.9.239 port 57340 [preauth]
2026-09-04 23:05:50.232	2026-09-04T22:05:50.232195+00:00 vps-le729caf sshd-session(42196): Invalid user pul from 66.93.9.239 port 57340
2026-09-04 23:05:48.991	2026-09-04T22:05:48.991572+00:00 vps-le729caf sshd-session(42193): Connection closed by invalid user solana 195.178.110.30 port 48342 [preauth]
2026-09-04 23:05:48.943	2026-09-04T22:05:48.943387+00:00 vps-le729caf sshd-session(42193): Invalid user solana from 195.178.110.30 port 48342
2026-09-04 23:04:24.242	2026-09-04T22:04:24.342227+00:00 vps-le729caf sshd-session(42190): Connection reset by authenticating user root 45.146.10.151 port 47834 [preauth]
2026-09-04 23:03:02.792	2026-09-04T22:03:02.792069+00:00 vps-le729caf sshd-session(42188): Connection closed by invalid user solana 195.178.110.30 port 42508 [preauth]
2026-09-04 23:03:02.749	2026-09-04T22:03:02.749609+00:00 vps-le729caf sshd-session(42188): Invalid user solana from 195.178.110.30 port 42508
2026-09-04 23:01:05.838	2026-09-04T22:01:05.838622+00:00 vps-le729caf sshd-session(42185): Connection closed by invalid user wpyan 66.93.9.239 port 52556 [preauth]
2026-09-04 23:01:05.738	2026-09-04T22:01:05.738104+00:00 vps-le729caf sshd-session(42185): Invalid user wpyan from 66.93.9.239 port 52556
2026-09-04 23:00:26.582	2026-09-04T22:00:26.582890+00:00 vps-le729caf sshd-session(42180): Connection closed by invalid user sol 195.178.110.30 port 37232 [preauth]
2026-09-04 23:00:26.465	2026-09-04T22:00:26.465589+00:00 vps-le729caf sshd-session(42180): Invalid user sol from 195.178.110.30 port 37232
2026-09-04 22:59:30.886	2026-09-04T21:59:30.866921+00:00 vps-le729caf sshd-session(42172): Connection closed by authenticating user root 138.226.239.233 port 24124 [preauth]
2026-09-04 22:57:16.472	2026-09-04T21:57:16.472882+00:00 vps-le729caf sshd-session(42170): Connection reset by authenticating user root 45.146.10.141 port 56114 [preauth]